

ALIF JAWAD

Tucson, Arizona, United States

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EDUCATION

University of Arizona

Doctor of Philosophy(PhD) - First Year Student of Mechanical Engineering

Current CGPA: 3.75/4.00

Research Supervisor: Dr. Majid Beidaghi

2024 – Present

Arizona, United States

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering

CGPA: 3.53/4.00

2018 – 2023

Dhaka, Bangladesh

Thesis: Analysis of Tensile Strength of High Entropy Alloy (FeCrCuCoNi) using a Combination of Molecular Dynamics Simulations and Machine Learning

RESEARCH INTERESTS

- MXene
- Molecular Dynamics
- Applied Machine Learning
- 2D Materials
- Material Characterization
- Material Informatics

RESEARCH EXPERIENCE

Undergraduate Thesis Supervisor:

Dr. Jane Alam Khan

Undergraduate Thesis: Analysis of Tensile Strength of High Entropy Alloys using a combination of Molecular Dynamics Simulations and Machine Learning

Summary:

- Finding the tensile Strength of HEA using MD simulations of 121 samples of single crystal as Low-Fidelity Data and 41 samples of polycrystal as High Fidelity Data.
- Fitting the generated data in MPINN model for predicting tensile strength under varying composition.
- Analyzing error to establish relation between high fidelity data and low fidelity data.

PhD Researcher, Beidaghi Electrochemistry Research Group

Supervisor: Dr. Majid Beidaghi

2024 – Present

University of Arizona

- Optimizing synthesis conditions for Ti, Nb, and V MAX phases using Flash Joule Heating.
- Exploring synthesis pathways from low-entropy to high-entropy MAX phases using Flash Joule Heating.
- Collaboration: Testing the electrochemical performance of Si with MXene ink as an electrode material.
- Collaboration: Preparation of delaminated MXene films with different intercalants for filtration system applications.

PUBLICATIONS

Journal of Molecular Modeling

2025

N. E. Chowdhury, A. Jawad, A. Rahman, M. J. A. Khan

- Multi-fidelity neural network-based prediction of tensile strength of high-entropy alloy (FeNiCoCrCu) using molecular dynamics data. *Journal of Molecular Modeling*, 31(8), 1–9.

TEACHING EXPERIENCE

Fundamentals of Materials for Engineers (AME-MSE 331R)

Fall 2024, Spring 2025, Fall 2025, Spring 2026

Course Instructor: Dr. Majid Beidaghi

University of Arizona

- Helped the grading of the course by grading during homeworks, projects and final assessments
- Conducted problem solving sessions for the students
- Assisted the students by solving individual problems both during office hours and online sessions
- Recorded videos and assisted the students for their final course project

SKILLS SUMMARY

Materials Characterization:	XRD, SEM, AFM, EDS, Raman Spectroscopy, UV-Vis Spectroscopy, TGA
Electrochemical Characterization:	Cyclic Voltammetry (CV), Galvanostatic Charge-Discharge (GCD), Electrochemical Impedance Spectroscopy (EIS)
Lab Equipment:	Furnace (solid-state synthesis), Flacktek Speed Mixer, Centrifuge, Glovebox, Fume Hood, Flash Joule Heating
Simulation Software:	LAMMPS, MATLAB, ANSYS Workbench
Visualization Software:	OVITO, Tecplot 360, Origin
Programming Languages:	Python (Machine Learning, Deep Learning), C, Arduino
CAD Software:	SolidWorks, AutoCAD
Office Applications:	Microsoft Office, Excel, PowerPoint
Soft Skills:	Management, Leadership, Teaching, Writing, Public Speaking

PROFESSIONAL EXPERIENCE

Industrial Attachment Trainee

2022

Unilever Bangladesh Limited

Chittagong, Bangladesh

- Calculated the cooling load of a production floor and some of the measured dimensions and calculations were incorporated into existing draft.
- Tabulated and optimized water loss in each process.

EXTRACURRICULAR ACTIVITIES

Ennovators 6.0 by Banglalink

2022

Top 140

- After few rigorous rounds of competition, reached the top 140 competitors nationally to compete in a business case competition.

Captain

2022

Hall Volleyball, Badminton, Basketball, Table-Tennis Teams

- Organized and managed the hall teams for different sports and we reached the semi-finals in most of them.
- Created hall funds for the advancement and nourishment of players.

Public Relations Officer

2025–2026

Bangladesh Students Association, University of Arizona

- Organized club events and coordinated activities for the Bangladeshi student community.
- Managed outreach and communications to increase engagement and participation in association programs.

REFERENCES

Dr. Jane Alam Khan

Assistant Professor,
Department of Mechanical Engineering,
Bangladesh University of Engineering and Technology.
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Dr. Mohammad A. Motalab

Professor,
Department of Mechanical Engineering,
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